

Q1. What is persistence of vision?

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A1. Persistence of vision is the phenomenon due to which an afterimage is thought to persist for approximately one-twenty-fifth of a second on the retina or brain. Scientists are still under debate on the real cause of this phenomenon. Some scientists believe that the human brain and/or eye retain images for a fraction of a second. That is why we do not completely lose the link from the world when we blink.

However, many scientists and psychologists don't agree with this theory as it lacks medical evidence. Whatever the reason, the phenomenon surely exists, making applications like motion pictures on television, static and scrolling text on LED displays, and propeller clocks possible.

All displays take advantage of this



Fig. 1: Persistence of vision (source: Wikipedia)



Fig. 2: LED displays (Source: Wikipedia)

phenomenon in some way or the other to display continuous text, graphics, images or video. Like in LED displays (Fig. 2), not all the LEDs glow at a time. The rows are continuously scanned at such a fast rate that LEDs appear to be glowing simultaneously. This means that only one row is activated at a time and LEDs corresponding to the data for each column will illuminate. The rows are activated cyclically and one by one in synchronisation with the data at a very fast rate, so that the human eyes perceive the displayed text as stable.

Q2. We are planning to implement some Open Source EDA for our design and R&D departments. Which software will suit our needs?

Pooja Juyal
Through e-mail

A2. This is a very big decision for an organisation to make and will have its impact on almost all the departments (R&D, design, testing, QA, etc). It is always recommended to use the same electronic design automation (EDA) tool throughout the organisation. This means that one should clearly understand the requirements of different departments before selecting an EDA tool. Implementation of a new EDA will take a lot of efforts and training, so you will have to do it right the first time.

I see two strong candidates as the Open Source EDA tool of choice: gEDA project and KiCad. Both the tools have their own advantages and disadvantages, but being Open Source, these can be configured to some extent as per your needs and you will never have to pay any money.

gEDA project. gEDA project is one of the most matured EDA tools that supports complete electronics design cycle, including schematic capture, attribute management, bill-of-materials

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